

Troubleshooting Guide for Bluetooth Device Error Codes BT-GM-3921 and BT-BP-2710

Category: Procedures

Model: Bluetooth-Error-Guide

Description: This troubleshooting guide details the meaning of specific ingestion error codes BT-GM-3921 and BT-BP-2710, typical causes, and step-by-step remediation strategies to handle missing fields and connection retries for glucometers and BP cuffs affected during the firmware rollout impact period.

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1. Introduction

This document provides a detailed troubleshooting framework for addressing Bluetooth error codes BT-GM-3921 and BT-BP-2710 encountered during device operation within clinical workflows. These error codes are associated with connectivity and data ingestion issues in glucometers (BT-GM series) and blood pressure cuffs (BT-BP series). The guidance herein is pertinent during the firmware impact window from September 1, 2025, to November 10, 2025, during which firmware rollouts and vendor updates have increased incident reports.

The goal of this guide is to enable clinical and technical staff to quickly diagnose, resolve, and prevent recurrence of these error conditions, ensuring continuous data flow and patient safety.

2. Overview of Error Codes

Bluetooth device error codes serve as diagnostic indicators pointing to specific issues in connectivity, data transmission, or device configuration. During the recent firmware update cycle, two error codes became predominant:

- **BT-GM-3921:** Indicates a missing or incomplete glucose reading data during ingestion.
- **BT-BP-2710:** Denotes a failed connection attempt or retry exhaustion for blood pressure cuffs.

These codes are elevated during device-integration periods when firmware incompatibilities, missing fields, or communication disruptions occur.

3. Device-Specific Error Codes

3.1 Glucometer Error Code: BT-GM-3921

Symptom: The device logs show incomplete or missing glucose

measurement data, preventing successful data upload to clinical systems.

Root Causes:

- Firmware incompatibility leading to partial data transmission.
- Transient Bluetooth connectivity issues during data transfer.
- Device misconfiguration post-update, especially missing mandatory fields.

Impact: Inability to record or upload glucose readings, potentially affecting patient management and clinical decision-making.

3.2 Blood Pressure Cuff Error Code: BT-BP-2710

Symptom: Repeated connection retries fail, with the device generating BT-BP-2710 error logs, indicating persistent communication issues.

Root Causes:

- Firmware versions incompatible with the device's Bluetooth stack.
- Obstructions or interference in the Bluetooth environment.
- Configuration errors with device pairing or authentication parameters.

Impact: Device remains unpaired or unresponsive, delaying blood pressure measurements and clinical assessments.

4. Common Causes of Error Codes During Firmware Impact Period

- **Firmware Rollout Incompatibilities:** Firmware update introduced changes that are not backward-compatible, leading to missing data fields or connection failures.
- **Configuration Mismatches:** Post-update device settings may have been altered, causing communication or ingestion errors.
- **Environmental Interference:** Increased Bluetooth traffic or devices operating on overlapping frequencies during rollout.
- **Device Defects or Hardware Failures:** Rarely, hardware issues triggered by firmware changes may contribute.

Understanding these root causes informs targeted remediation strategies.

5. Remediation Procedures Summary

1. Verify device firmware versions and compatibility.
2. Inspect device configuration settings for completeness and correctness.
3. Check Bluetooth environment for interference or obstructions.
4. Perform targeted resetting and re-pairing of affected devices.
5. Collect detailed logs and error reports for further analysis.
6. Coordinate with vendor support for exchange if hardware defects are suspected.

Following these procedures systematically reduces downtime and restores data integrity.

6. Step-by-Step Troubleshooting Workflow

6.1 Preliminary Verification

- Confirm the exact error code displayed in device logs and clinical systems.
- Check device firmware version against vendor bulletin DEVCO-GM v3.9 for glucometers and DEVCO-BP v2.7 for BP cuffs.
- Ensure devices are powered on, within Bluetooth range, and have sufficient battery life.

6.2 Verify Firmware and Settings

- Navigate device menus or firmware management tools to verify firmware version.
- Compare current device settings against baseline configurations documented in deployment protocols.
- If firmware is outdated or incompatible, proceed with firmware reinstallation or update.

6.3 Network and Environment Check

- Scan for Bluetooth interference sources; switch to a less congested channel if accessible.
- Ensure no physical obstructions between device and receiver.
- Review device pairing history and retry connection procedures.

6.4 Reset and Re-Pairing

1. Perform a soft reset on the device following manufacturer instructions.
2. Remove existing pairing profiles on both device and host system.
3. Re-initiate device pairing, ensuring proper authentication exchanges.
4. Conduct a test measurement to verify successful connection and data ingestion.

6.5 Data Capture and Logging

- Collect logs immediately after troubleshooting attempts for analysis.
- Record error occurrences, timestamps, and remedial actions taken.
- Utilize vendor-specific diagnostic tools, if available, for detailed analysis.

6.6 Confirm Resolution

- Verify that subsequent data transmissions are complete and error-free.
- Document successful resolution steps and update troubleshooting records.
- If issues persist, escalate to vendor support with gathered logs and images.

7. Error Code Specific Guidance

7.1 Handling BT-GM-3921 (Missing Data)

- **Symptoms:** Missing glucose values in ingestion logs, failed uploads.
- **Resolution Steps:**
 1. Check for recent firmware updates aligning with vendor

bulletins.

2. Ensure calibration and configuration parameters include all required data fields.
3. Run a full device reset and re-enter configuration settings.
4. Test data ingestion post-reset with a controlled measurement.

- **Prevention:** Regular firmware validation and configuration audits during updates.

7.2 Handling BT-BP-2710 (Connection Retry Failures)

- **Symptoms:** Persistent connection attempt logs, no data transfer.

- **Resolution Steps:**

1. Ensure homologated firmware version compatibility.
2. Verify Bluetooth pairing credentials and device visibility states.
3. Conduct interference mitigation measures (move away from other wireless devices).
4. Test with alternative Bluetooth adapters or access points if possible.
5. Perform device firmware reinstallation per vendor instructions.

- **Prevention:** Scheduled firmware validation and environment audits post rollout.

8. Clinical Impact and Considerations

The presence of error codes BT-GM-3921 and BT-BP-2710 directly affects clinical workflows by delaying critical data entry and patient assessments. Ingestion failures of glucose readings hinder diabetes management, while connection failures for BP cuffs compromise cardiovascular monitoring accuracy.

Timely troubleshooting reduces potential risks, such as misinterpretation of patient data, delayed medications, or misdiagnoses. Continuous monitoring and proactive device management during firmware periods are essential for patient safety.

9. Preventive Measures and Best Practices

- Implement routine firmware and configuration audits aligned with vendor patch releases.
- Establish environmental controls to minimize Bluetooth interference during rollout periods.
- Train clinical staff on basic troubleshooting and device reset procedures.
- Maintain updated device logs and documentation for rapid diagnosis.
- Coordinate closely with vendor support for timely firmware updates and known issues communication.

Adopting these practices enhances robustness and minimizes workflow disruptions.

10. Appendices

10.1 Vendor Bulletins

Document	Date	Description
DEVCO-GM v3.9	2025-10-15	Glucose meter firmware update addressing ingestion errors including BT-GM-3921.
DEVCO-BP v2.7	2025-09-28	Blood pressure cuff firmware upgrade resolving BT-BP-2710 connection issues.

10.2 Clinician Outreach Scripts

Sample Script:

"Hello, this is a support follow-up regarding your Bluetooth device issue. We are aware of intermittent connection problems related to recent firmware updates. Please ensure your device firmware is up-to-date (v3.9 for glucometers, v2.7 for BP cuffs). Try resetting the device and re-

pairing it following the steps in the troubleshooting guide. If issues persist, contact support with logs."

10.3 Next Best Action Policy Thresholds

Parameter	Threshold	Action
Fasting Glucose	> 140 mg/dL	Trigger nurse check-in and repeat measurement within 1 hour.
Blood Pressure	> 180/120 mmHg	Immediate alert to clinician; schedule urgent review.

10.4 Risk Stratification Model Card - Feature Importance

Feature	Importance Score	Description
Device Firmware Version	0.85	Higher scores indicate risk during firmware rollout due to incompatibility.
Bluetooth Signal Strength	0.65	Weaker signals correlate with higher error likelihood.

10.5 Clinical Note Generator QA Guidance

- Ensure all device error logs are accurately documented with timestamps.
- Confirm that troubleshooting steps are clearly recorded and include resolution confirmation.
- Validate that any clinical decisions based on device data are supported by proper documentation.
- Regularly review QA metrics on device error frequency and resolution success rate.